

Purpose of This Guide

This guide provides a step-by-step, end-to-end walkthrough of the Secure Exchange Process used to create verification units for TrustFLEX and TrustCUSTOM secure ICs. The process ensures the secure transfer of customer-defined device configuration and cryptographic key material to Microchip for controlled provisioning and manufacturing. The instructions in this guide are based on Microchip's official Secure Exchange Process tutorial and are applicable across supported secure IC devices.

When to Use the Secure Exchange Process

Use the Secure Exchange Process after prototyping is complete and you are ready to:

- Lock down device configuration
- Inject real customer cryptographic key secrets
- Order verification units prior to production

The verification unit phase enables validation of the final configuration before authorizing full production.

Prerequisites

Before starting the Secure Exchange Process, ensure you have:

- A myMicrochip account
- Trust Platform Design Suite (TPDS) installed
- Completed prototyping using dummy keys only
- Finalized use cases and device configuration

High-Level Workflow Overview

1. Define final security use cases and configuration
2. Complete the Secure Exchange Questionnaire
3. Submit the questionnaire to Microchip via myMicrochip support case
4. Receive customer-specific RSA public key (and optional CSR)
5. Generate encrypted provisioning package using TPDS
6. Submit the encrypted provisioning package to Microchip
7. Order verification units via MicrochipDIRECT
8. Test verification units
9. Approve the configuration for production

Each step is described in detail in the sections that follow.

Figure 1. Secure Exchange Process Path to Production

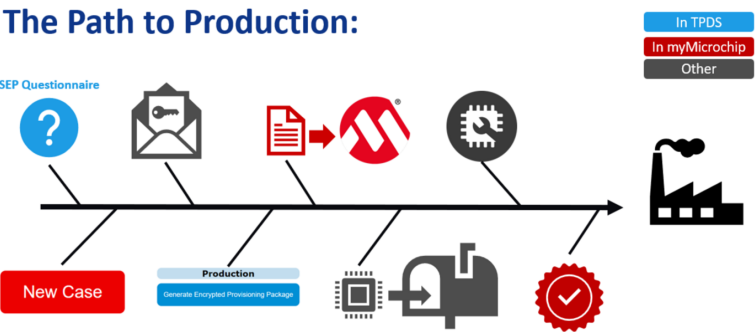


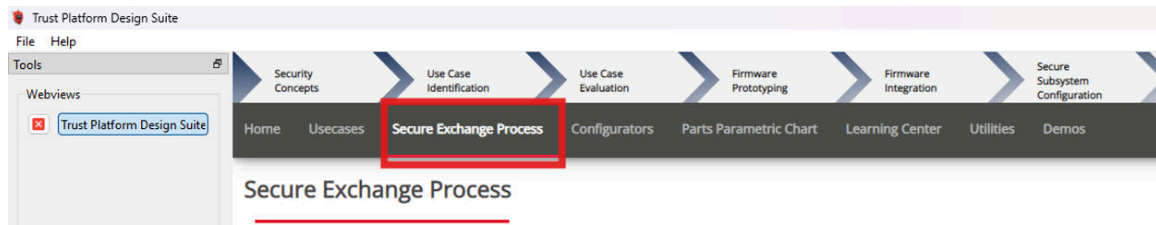
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1. Complete the Secure Exchange Questionnaire

1. Launch the Trust Platform Design Suite.
2. Navigate to the Secure Exchange Process tab.

Figure 1-1. TPDS Secure Exchange Process Tab



3. Under Solution:
 - a. Select the secure IC (e.g., ATECC608)
 - b. Select the provisioning flow (TrustFLEX or TrustCUSTOM)
 - c. Select the package (e.g., SOIC-8)
 - d. Select the communication protocol (e.g., I²C)

Figure 1-2. Questionnaire Solution Section

4. Under Project:
 - a. Enter the Program Name
 - b. Enter the Program Version
 - c. Provide a Program Description
 - d. Specify the production timeframe
 - e. Provide one to three MicrochipDIRECT-registered email addresses

Figure 1-3. Questionnaire Project Section

5. Under Project Requirements (optional):
 - a. Indicate whether a custom PKI is required
 - b. Indicate whether WPC Qi 1.3 provisioning is required

Figure 1-4. Questionnaire Project Requirement Section

The screenshot shows the 'Project Requirements' section of a questionnaire. It features a dropdown menu labeled 'Project Requirements' with a downward arrow. Below the dropdown, the text 'Certificate Options:' is followed by three radio button options: 'No Certificates' (which is selected), 'MCHP Standard Certificate(MCHP as Root CA)', and 'Custom PKI(Customer as root CA)'.

6. Under Project Environment:
 - a. Select the main controller type
 - b. Identify controller vendors
 - c. Specify communication link (embedded system ↔ external system)

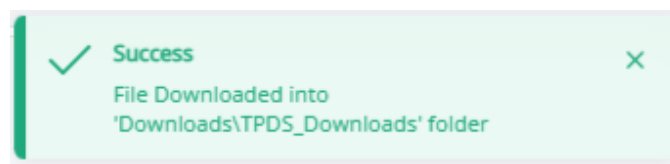
Figure 1-5. Questionnaire Project Environment Section

The screenshot shows the 'Project Environment' section of a questionnaire. It features a dropdown menu labeled 'Project Environment' with a downward arrow. Below the dropdown, there are three sections of options:

- Main controller type of your embedded system:** Five radio button options: '8-bit', '16-bit', '32-bit', 'Microprocessor', and 'FPGA'.
- Vendors:** Two radio button options: 'Microchip' and 'Other vendor'.
- Communication link between your embedded system and the external system:** Seven checkbox options: 'Wifi', 'Bluetooth / BLE', 'Cellular', 'CAN', 'Ethernet', 'USB', and 'Other'.

7. Select Save User Data.
 - a. This action downloads the completed Secure Exchange Questionnaire file

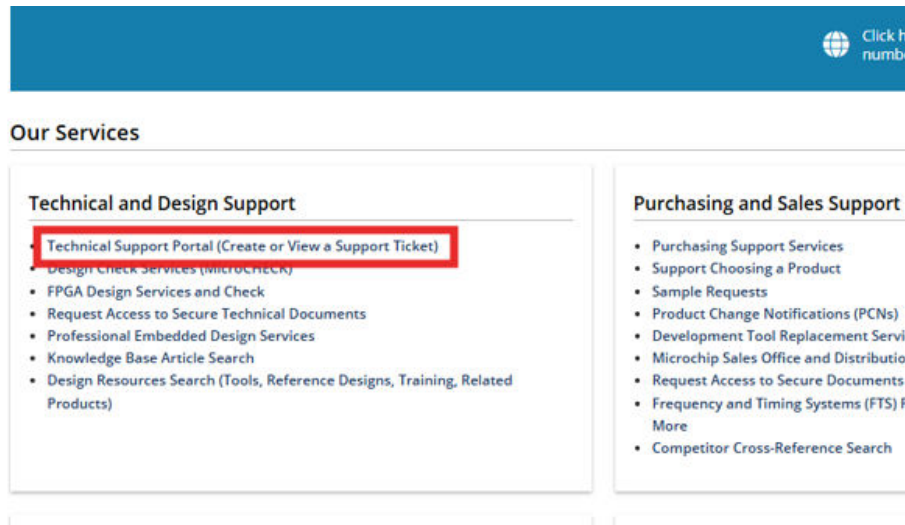
Figure 1-6. Download Success Message



2. Create a myMicrochip Support Case

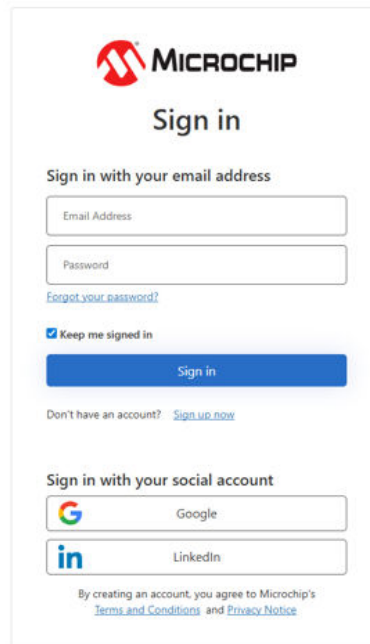
1. Go to [Technical Support](#) on Microchip.com.

Figure 2-1. Open a myMicrochip Support Case



2. Create a new myMicrochip account or log in to an existing account.

Figure 2-2. Log into myMicrochip



3. In the "Let us know how we can help you" section select:
 - a. Value-Added Services
4. Click Next.

Figure 2-3. Select Value Added Services

Let us know how we can help you

Case Reason*

- ☐ **Hardware/Firmware Support**
(Device, Peripherals, Modules, Code examples, Demo boards, Evaluation kits.)
- ☐ **Software Library**
(Harmony, Touch Library, Advanced Software Framework (ASF), Atmel Start, MPLAB® Code Configurator, Microchip Library of Applications, Bootloaders, Baremetal Softpack, Linux OS and drivers)
- ☐ **Development Tools**
(Development Environment - IDE, Compilers, Debugger, Emulator, Programmer, MPLAB Analog Designer, MPLAB Mindi Simulator, SW License, IP Core/IP License)
- ☐ **Documentation**
(Datasheet, Errata, Application Notes, FRM, EOL, Soldering, Packaging, demo/kit design files)
- ☐ **Quality/Reliability**
(If requesting FA/RMA for product that was purchased through a distributor, please contact the distributor directly. FIT/MTBF, MSI, RMA, FA, EOL, Part Marking. NOTE: For FPGA Product RMA/FA, select Hardware/Firmware Support.)
- ☐ **Product Selection**
(Help to select suitable product)
- ☒ **Value Added Services**
(Design Check, Trust Flex, Trust Custom)
- ☐ **Website Issues**
(Support website related issues.)

Next Cancel

5. In the "Provide more specific information" section, enter the following:
 - a. Subject: Provisioning the secure IC in the primary target device
 - b. Device: Secure IC (plain device number)
 - c. Additional Microchip Devices: Optional
 - d. Non-Microchip Devices Involved: Optional
 - e. Category: Provisioning Services
 - f. Subcategory: Provisioning flow selected in the questionnaire (TrustFLEX or TrustCUSTOM)
6. Click Next.

Figure 2-4. Device and Provisioning Flow Selection

Provide more specific information

Subject*
ATECC608 Provisioning

Primary Target Device*
ATECC608C

Additional Microchip Devices
search..

Non-Microchip Devices Involved

Category*
Provisioning Services

Sub-Category*
TrustFlex

Previous Next Cancel

7. In the "Describe your issue here" section, provide:
 - a. Project details
 - b. Devices to be provisioned
 - c. End-application description
8. Click Submit and Add Attachments.

Figure 2-5. Input Application Details

Describe your issue here

For Secure Provisioning Requests please download the Microchip Trust Platform Design Suite and fill out the Secure Exchange Process (SEP) Questionnaire within TPDS. Once completed, upload the saved form from TPDS to the support ticket so that we can begin secure provisioning efforts. If there are additional details or requirements that are part of your request, please include them as a comment when you upload the questionnaire.

Please add your issue description here*
Attachments can be added after submitting the Case

Describe use case for product and issue to be solved

Please enter Issue Description:
31930 characters remaining

Project Name / Registration Number
Project Name

Application Details
To help you with a prompt resolution, please tell us a bit more about the Application and End Customer; such information helps us route the case to the appropriate expert.

My Application details

226 characters remaining

Application Design Stage*
Development

Urgency*
Moderate

End Customer
(applicable only for Distri customers)

Previous Submit and Attach Files Cancel

9. Attach the downloaded Secure Exchange Questionnaire and click Done.

Figure 2-6. Attach Questionnaire File

Attach Files

Case has been successfully submitted, please add your attachments.

Do not upload any file that is Controlled, Classified, and/or require Export Authorization. If you are unsure, STOP and first contact your local Microchip sales representative.

Upload files Or drop files

SEPQuestionnaire_Userdata_0601132716

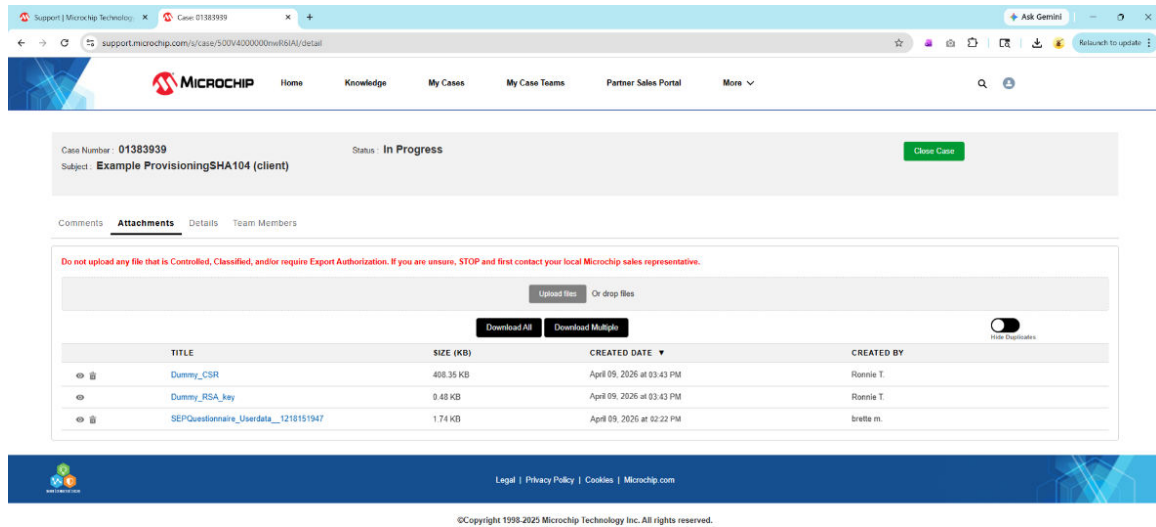
Remove

Done Skip

3. Generate the Encrypted Provisioning Package

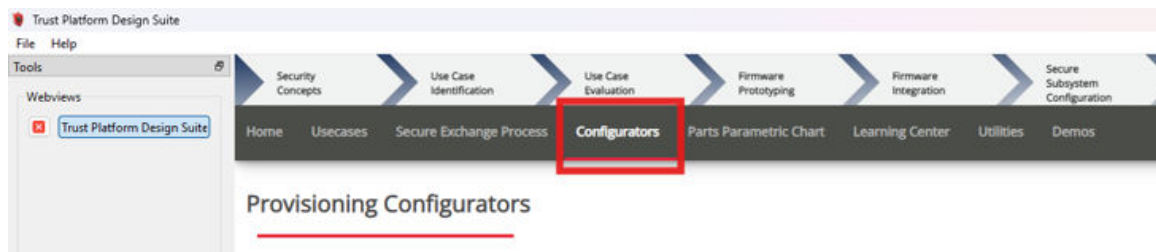
1. After reviewing your submission, Microchip will respond via your support case with the following:
 - a. An RSA public key (always provided)
 - b. A Certificate Signing Request (CSR) (*only if using a custom PKI*)
2. Download these files from the attachments section of the support case.

Figure 3-1. Download RSA Key and CSR



3. Return to the Trust Platform Design Suite.
4. Open the appropriate TrustFLEX/TrustCUSTOM configurator:

Figure 3-2. TPDS Configurator Tab

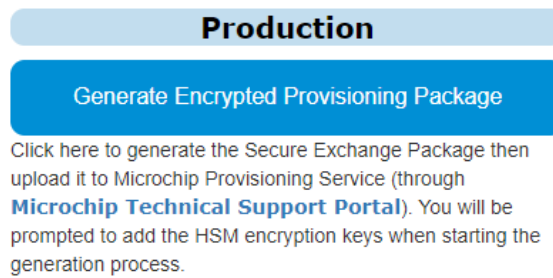


5. Configure the device exactly as intended for production, including:
 - a. Use cases
 - b. Keys
 - c. Certificates
 - d. Slot configuration

➔ Important: When generating the Encrypted Provisioning Package to request verification units, use real project secret keys for production so the final configuration and keys can be fully validated. During prototyping, use dummy keys to protect your real project secret keys.

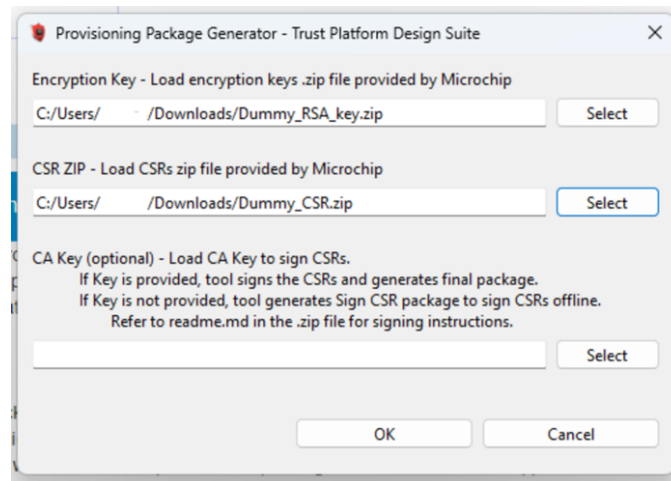
6. Click Generate Encrypted Provisioning Package.

Figure 3-3. Generate Encrypted Provisioning Package



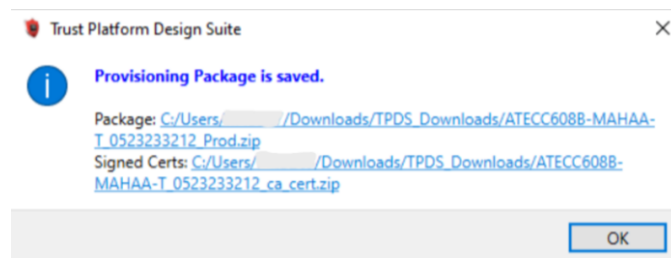
7. In the pop-up window, select the following files:
 - a. RSA key (required)
 - b. CSR (if applicable)
 - c. CA key (optional):

Figure 3-4. Upload RSA Key and CSR



8. Confirm the selection and download the encrypted provisioning package:

Figure 3-5. Download Confirmation



9. Return to your myMicrochip support case.
10. Under Attachments, click Upload Files.
11. Select and open the encrypted provisioning package.
12. Click Done.

Figure 3-6. Upload Encrypted Provisioning Package

The screenshot shows the Microchip support case interface. The case number is 01383939 and the status is In Progress. The subject is Example ProvisioningSHA104 (client). The Attachments tab is active, showing a table of uploaded files. The table has columns for Title, Size (KB), Created Date, and Created By. There are four files listed: Encrypted_Provisioning_Package (687.47 KB), Dummy_CSR (408.35 KB), Dummy_RSA_key (9.48 KB), and SEPQuestionnaire_Userdata_1218151947 (1.74 KB). All files were created on April 09, 2026 at 03:43 PM, except for the last one which was created at 02:22 PM. The created by for all files is brette m.

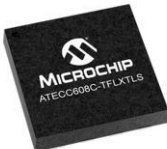
TITLE	SIZE (KB)	CREATED DATE	CREATED BY
Encrypted_Provisioning_Package	687.47 KB	April 09, 2026 at 03:43 PM	brette m.
Dummy_CSR	408.35 KB	April 09, 2026 at 03:43 PM	Ronnie T.
Dummy_RSA_key	9.48 KB	April 09, 2026 at 03:43 PM	Ronnie T.
SEPQuestionnaire_Userdata_1218151947	1.74 KB	April 09, 2026 at 02:22 PM	brette m.

4. Order Verification Units

After processing is complete, Microchip will provide instructions through your support case to proceed as follows:

- 1. In MicrochipDIRECT, navigate to the unique part number provided in the support case.
- 2. Click Place Verification Order.

Figure 4-1. Place Verification Order



Part Number: ATECC608C-TFLXTL9U
Lead Count: 8
Package Type: UDFN
Temp. Range: -40C to +85C
Packing Media: T/R

Program Name/Version: Microchip Demo / V1.0
Your Part Number: ATECC608C-MAHXX-T

Device Pricing	Comments	
Quantity		per Unit (\$USD)
1+		0.87
Large-Quantity Price Request available for these quantities or higher. Request a Price Quote.		

In Stock Now: 66,116

📦 Order now, up to 60,116 estimated to ship on 11-May-2024
📦 Order now, up to 6,000 estimated to ship on 10-May-2024

Lead Time For Additional Quantities. Order Now to Secure **Additional quantities** estimated to ship by 27-Jun-2024

Delivery and scheduling options available in the cart

Place Verification Order

Approve

Reject

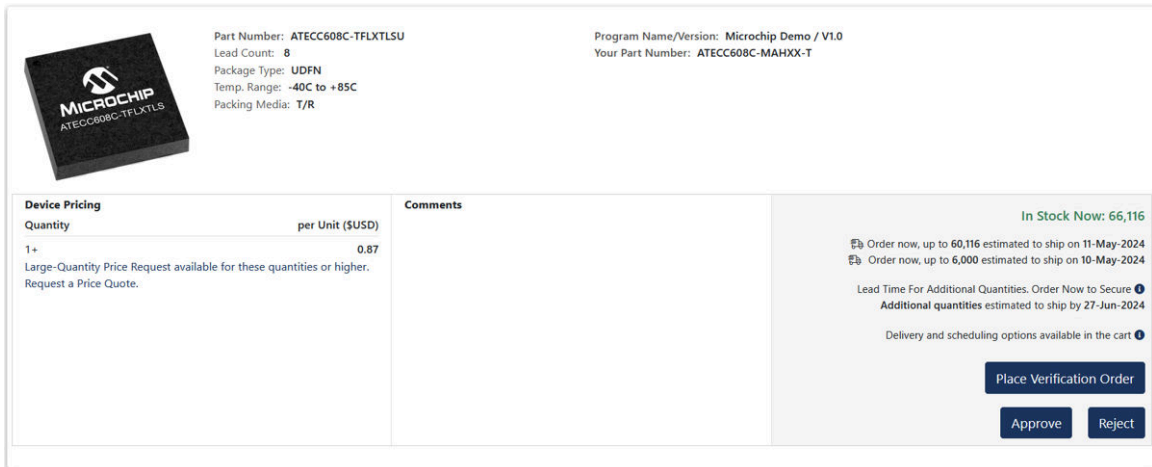
5. Approve Verification Units for Production

1. When verification units are received:
 - a. Test authentication flows
 - b. Validate certificates and keys
 - c. Confirm that device behavior matches expectations

 **Important:** This is the final validation checkpoint before production.

2. Return to the custom part number in MicrochipDIRECT.
3. Based on the validation results, select Approve or Reject.

Figure 5-1. Approve or Reject Verification Units



Part Number: ATECC608C-TFLX-TLSU
 Lead Count: 8
 Package Type: UDFN
 Temp. Range: -40C to +85C
 Packing Media: T/R

Program Name/Version: Microchip Demo / V1.0
 Your Part Number: ATECC608C-MAHXX-T

Device Pricing	Comments
Quantity per Unit (\$USD) 1+ 0.87 Large-Quantity Price Request available for these quantities or higher. Request a Price Quote.	

In Stock Now: 66,116

Order now, up to 60,116 estimated to ship on 11-May-2024
 Order now, up to 6,000 estimated to ship on 10-May-2024

Lead Time For Additional Quantities. Order Now to Secure Additional quantities estimated to ship by 27-Jun-2024

Delivery and scheduling options available in the cart

Place Verification Order

Approve Reject

Approval authorizes Microchip to use your secure exchange data and begin full production provisioning.

Rejection requires the customer to add a comment to the support ticket stating the reason. If configuration or functionality changes are required, a new support case must be submitted. Once the issue is resolved, the MicrochipDIRECT project will be re-enabled to place an order for new verification units.

6. Final Steps and Resources

Security Model Summary

- Dummy secrets are used for prototyping, while real customer secrets are used only for verification units and production
- Real customer secrets are protected using RSA-based encryption and are never transmitted in plaintext
- Microchip provisions verification devices within a secure environment using encrypted inputs and HSM-backed infrastructure

This model supports secure manufacturing at scale while maintaining customer ownership of cryptographic assets.

Next Steps

- Proceed to production ramp
- Contact Microchip Support for configuration changes or revisions

7. Revision History

Revision	Date	Description
A	06/2026	Initial release of this document

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